

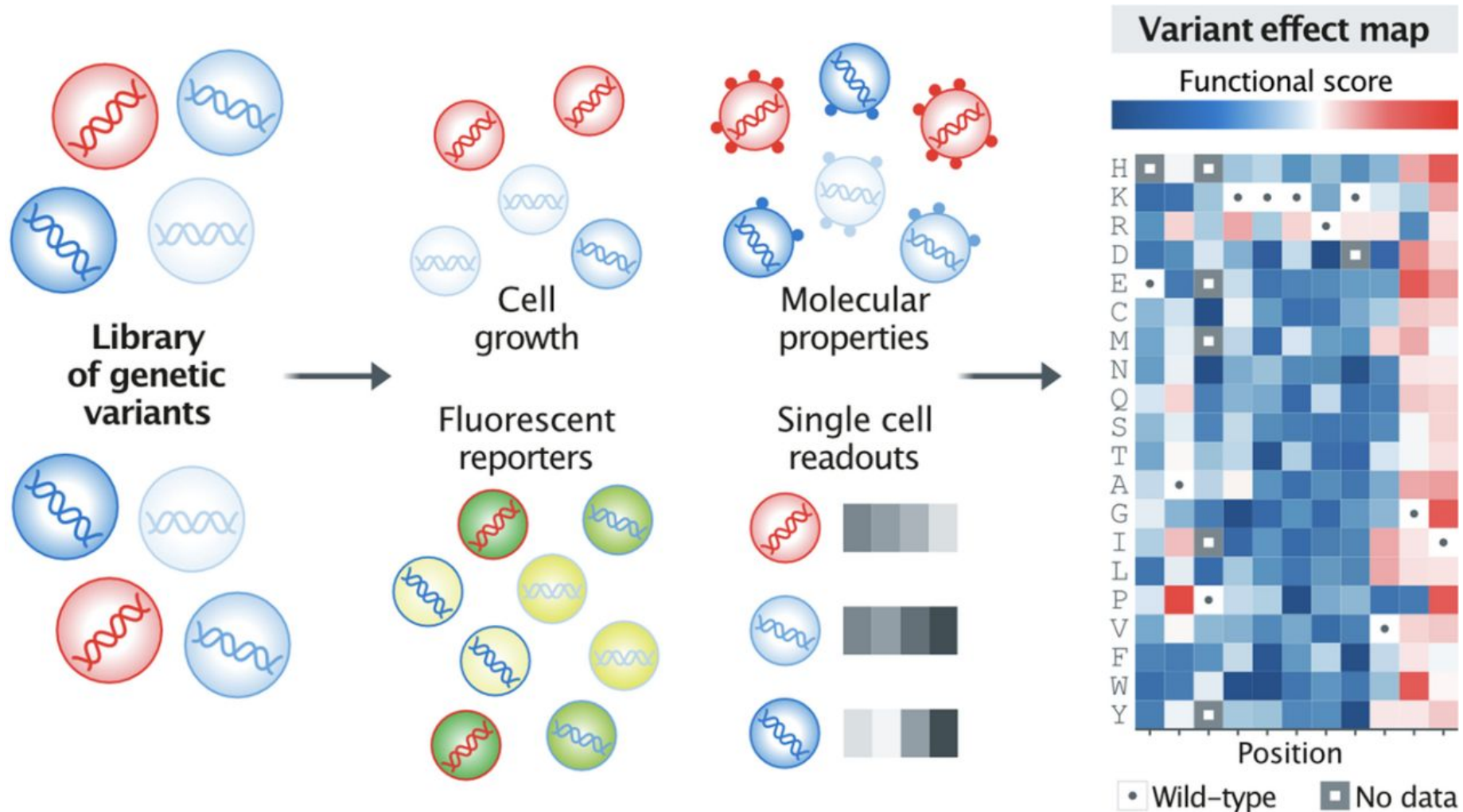
connecting people with science

Dr Alan Rubin

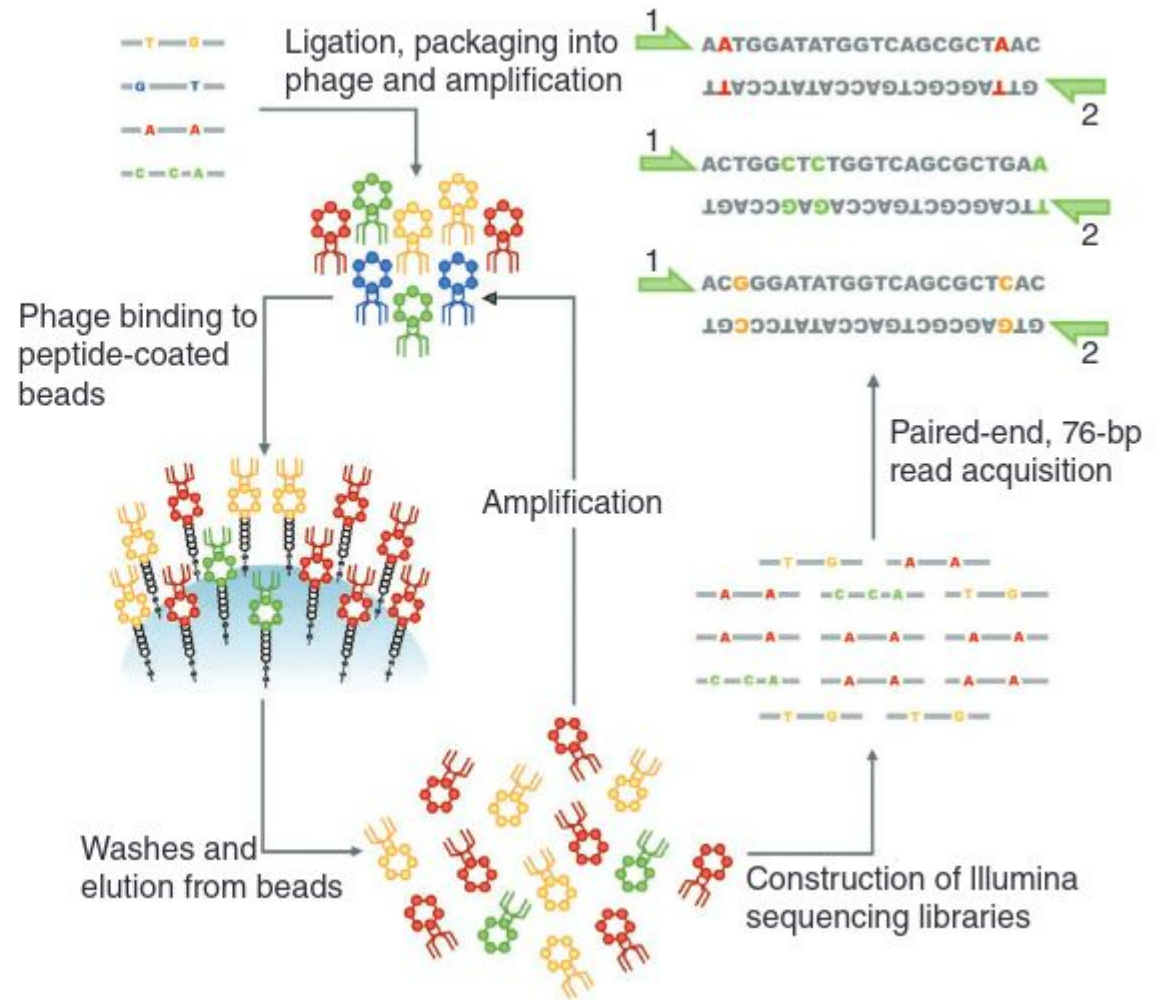
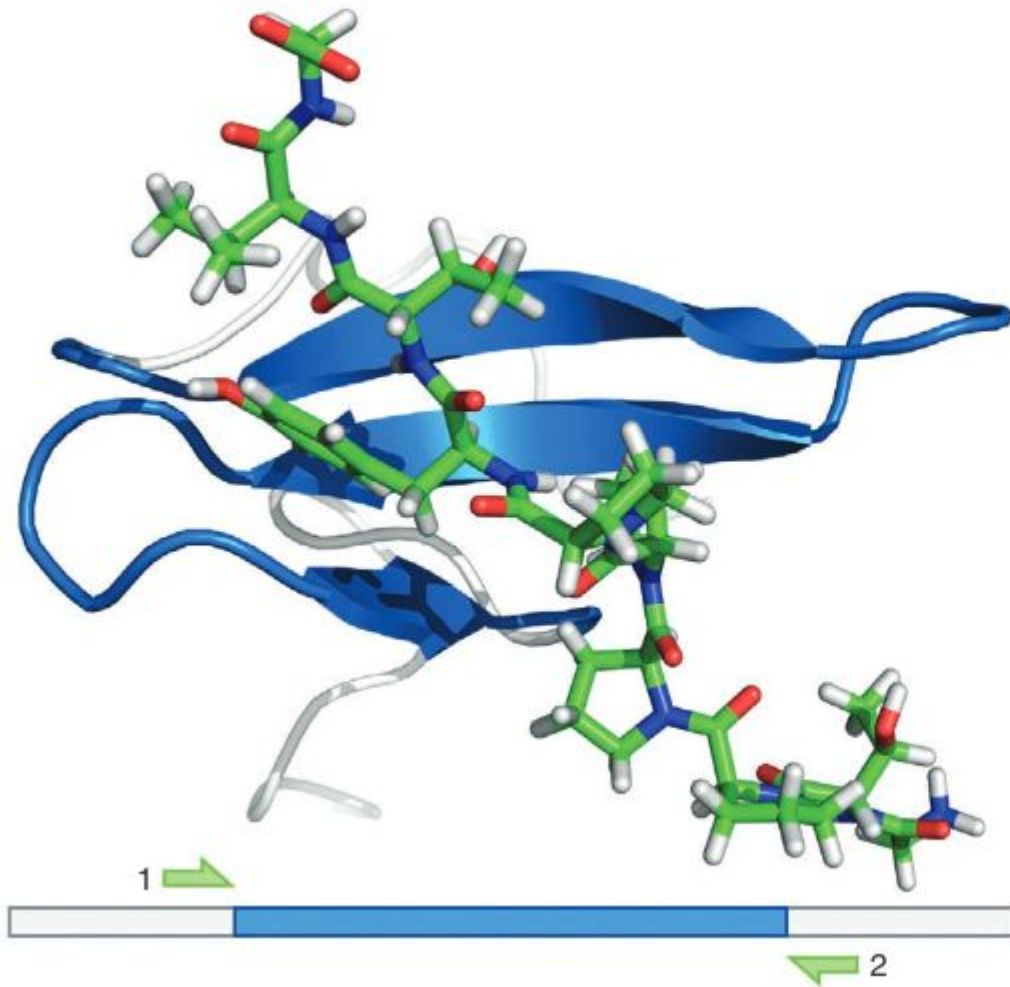
Data integration II
MaveDB

November 2025

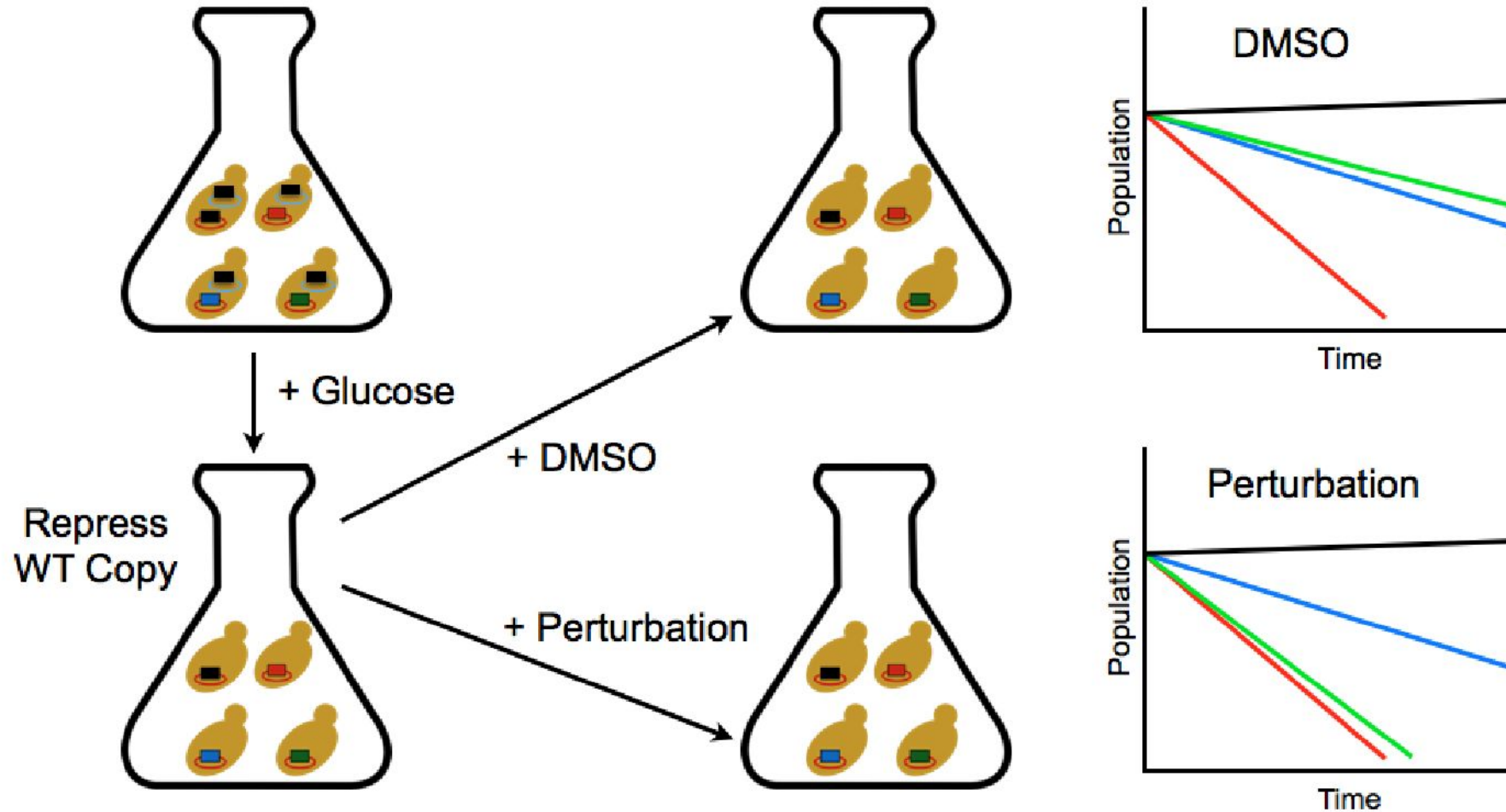
Multiplexed assays of variant effect



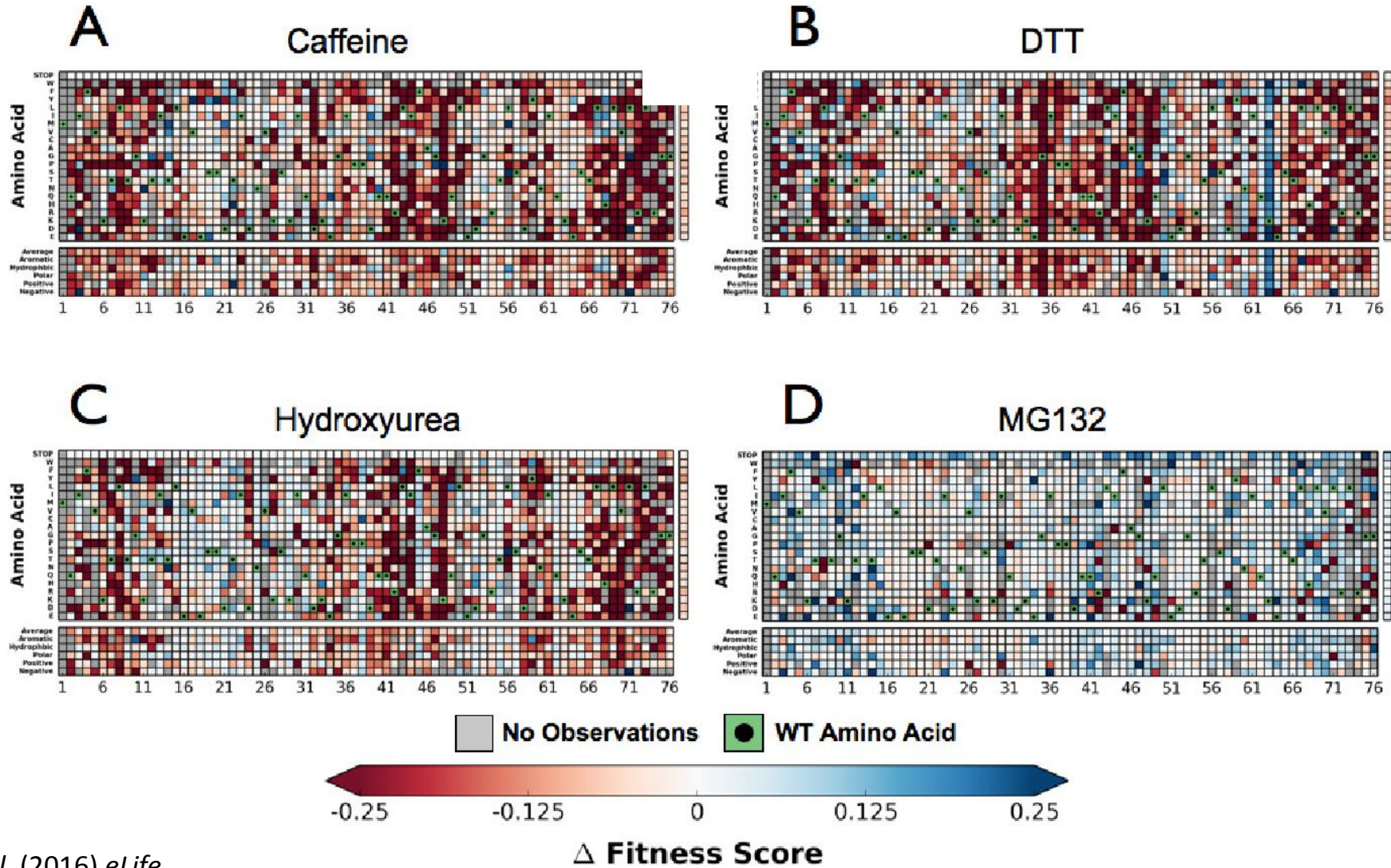
The original DMS: phage display



Yeast-based growth assays

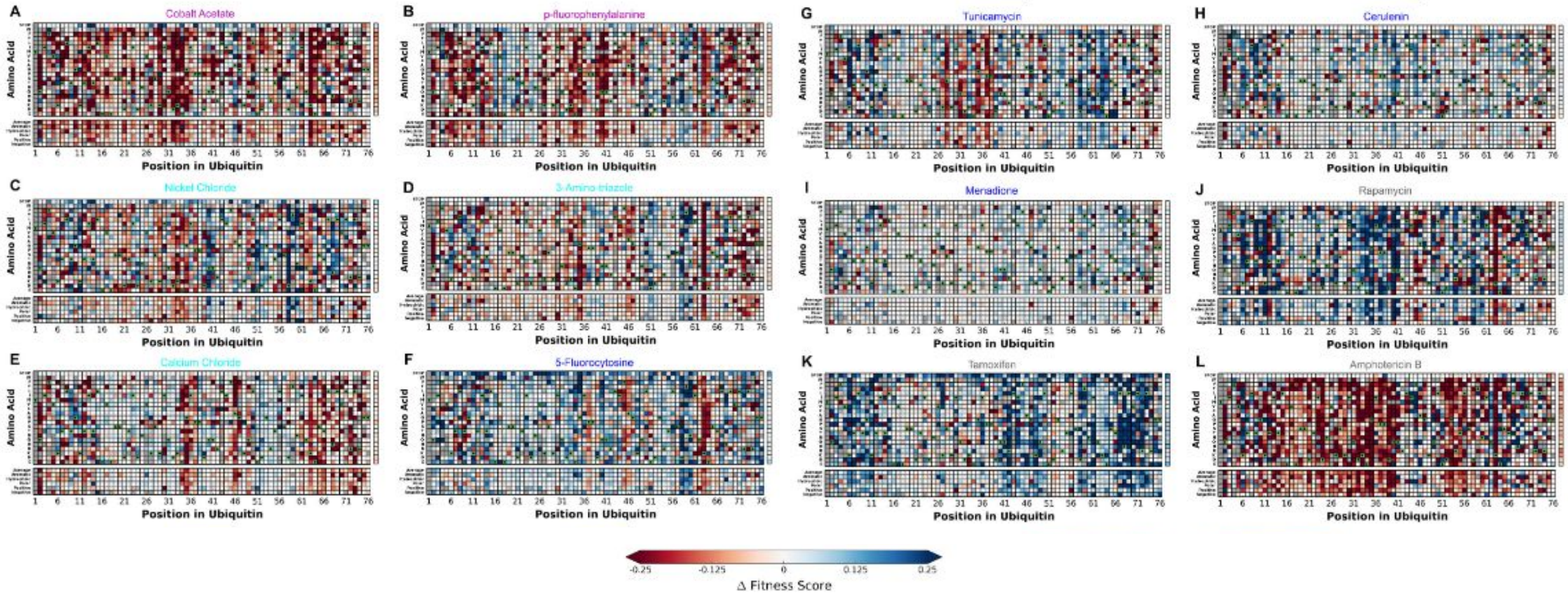


Yeast-based growth assays

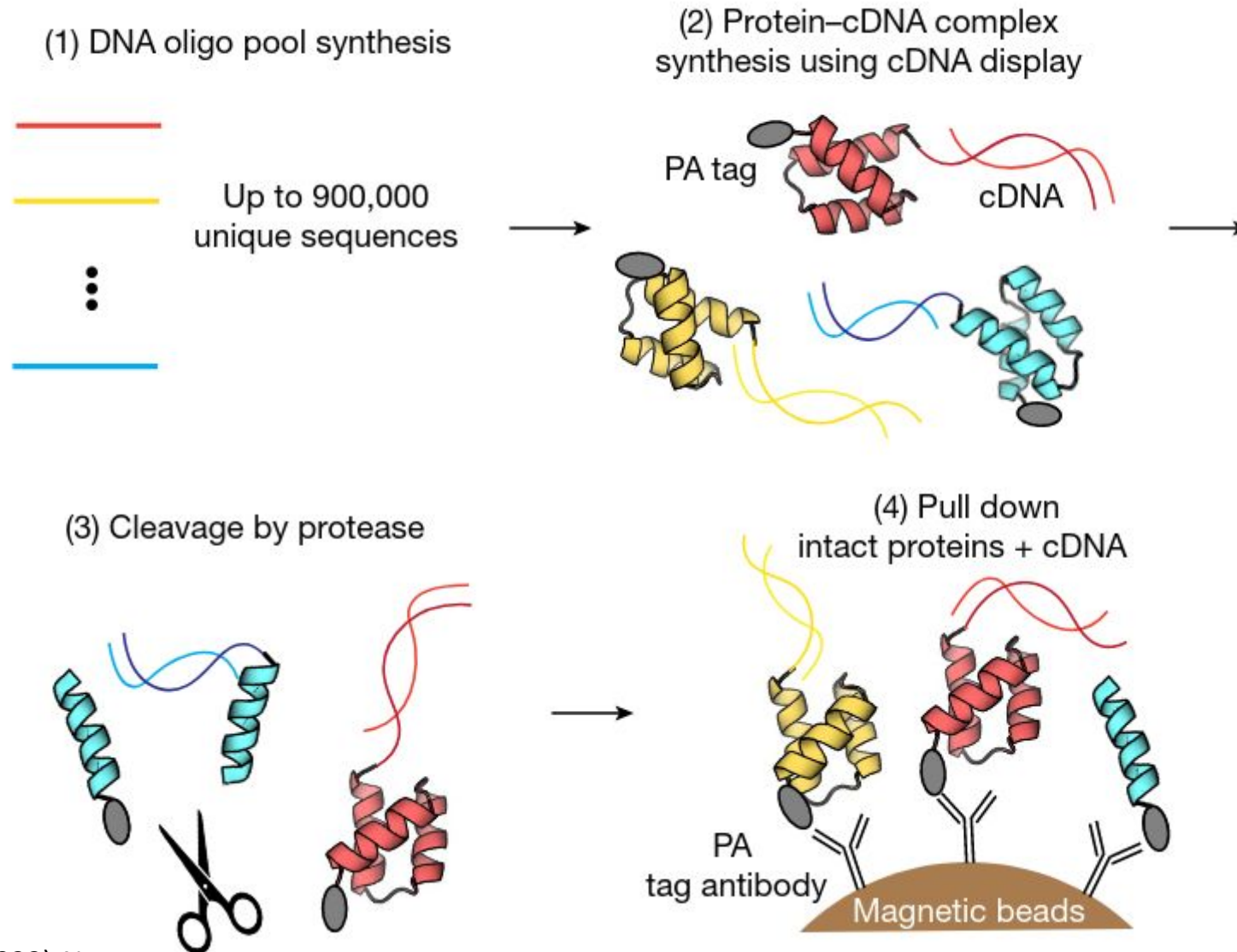


Adapted from Mavor *et al.* (2016) *eLife*.

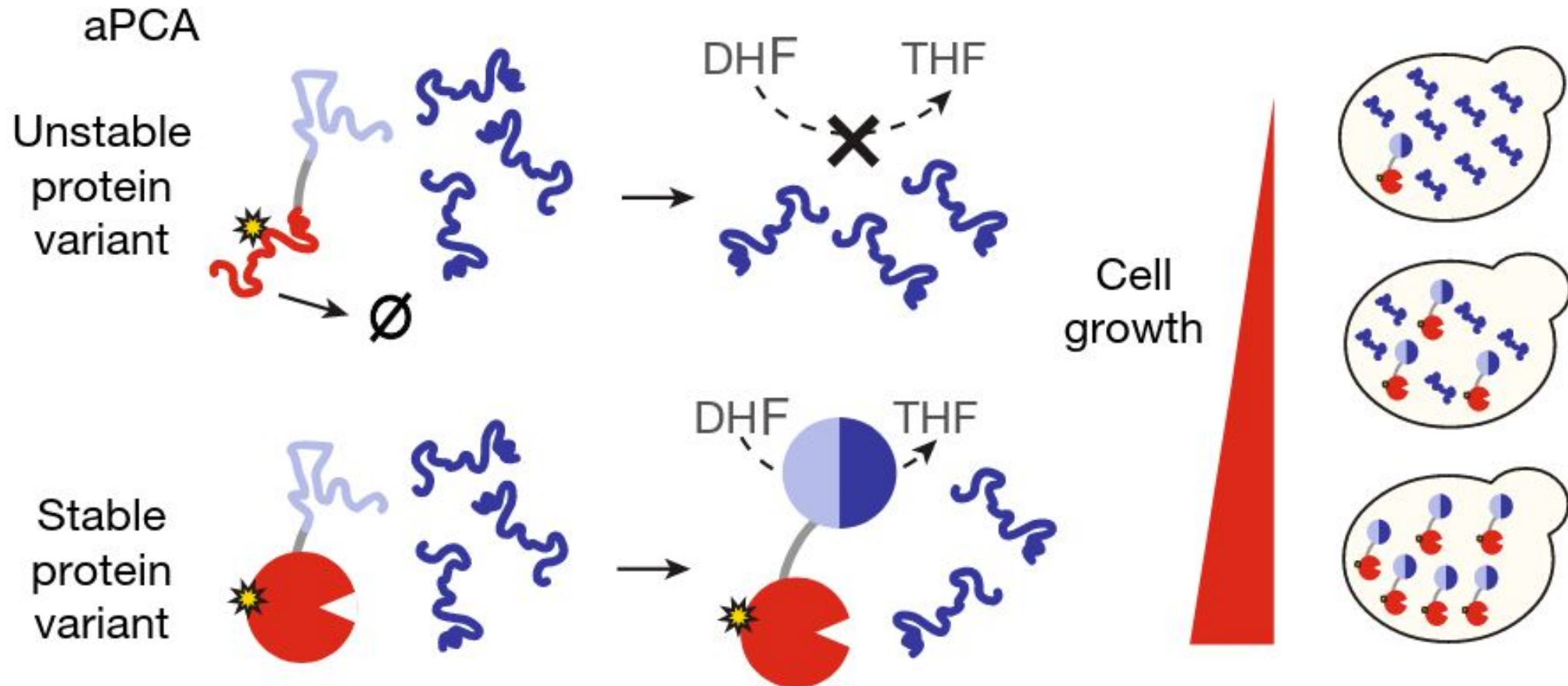
Yeast-based growth assays



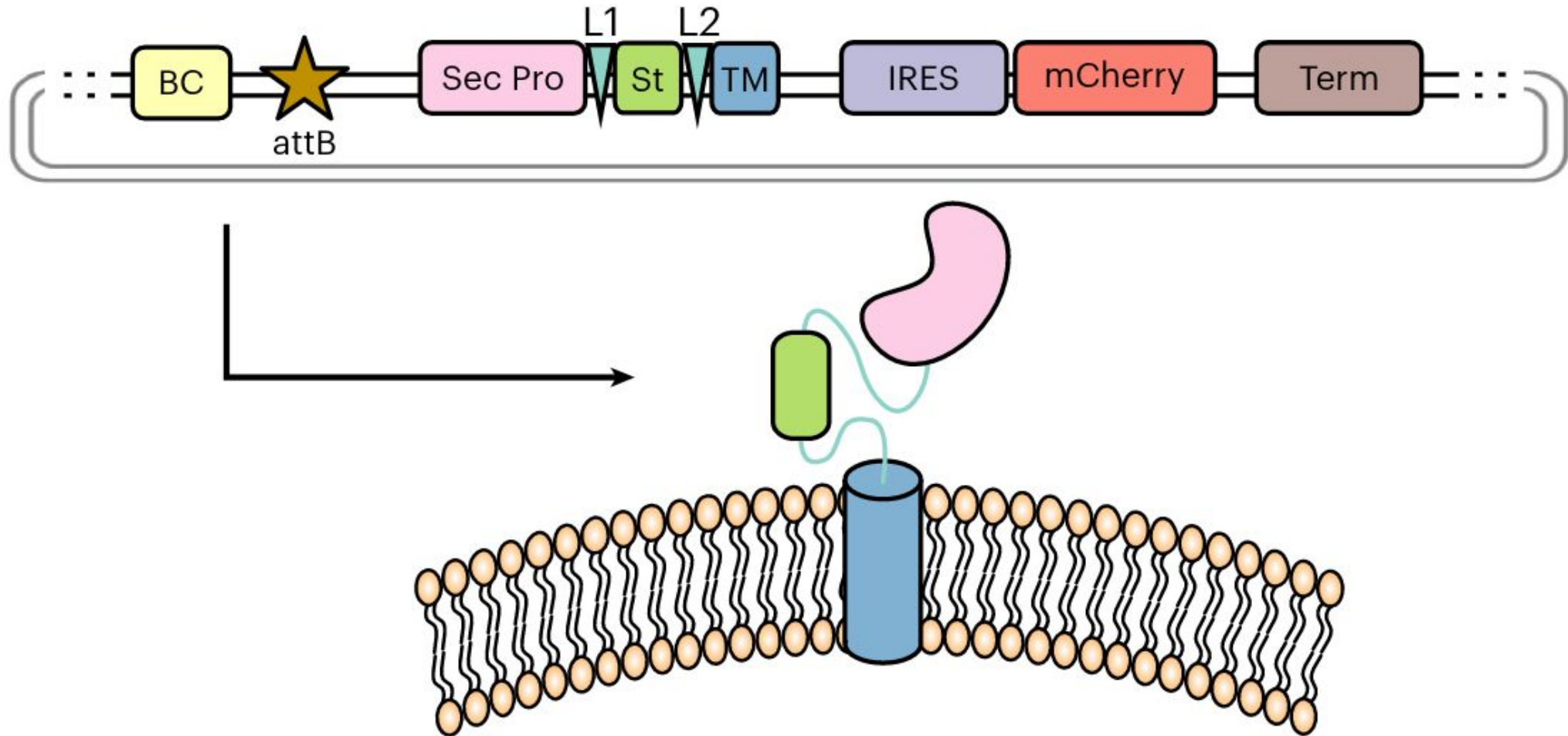
Mega-scale stability experiment



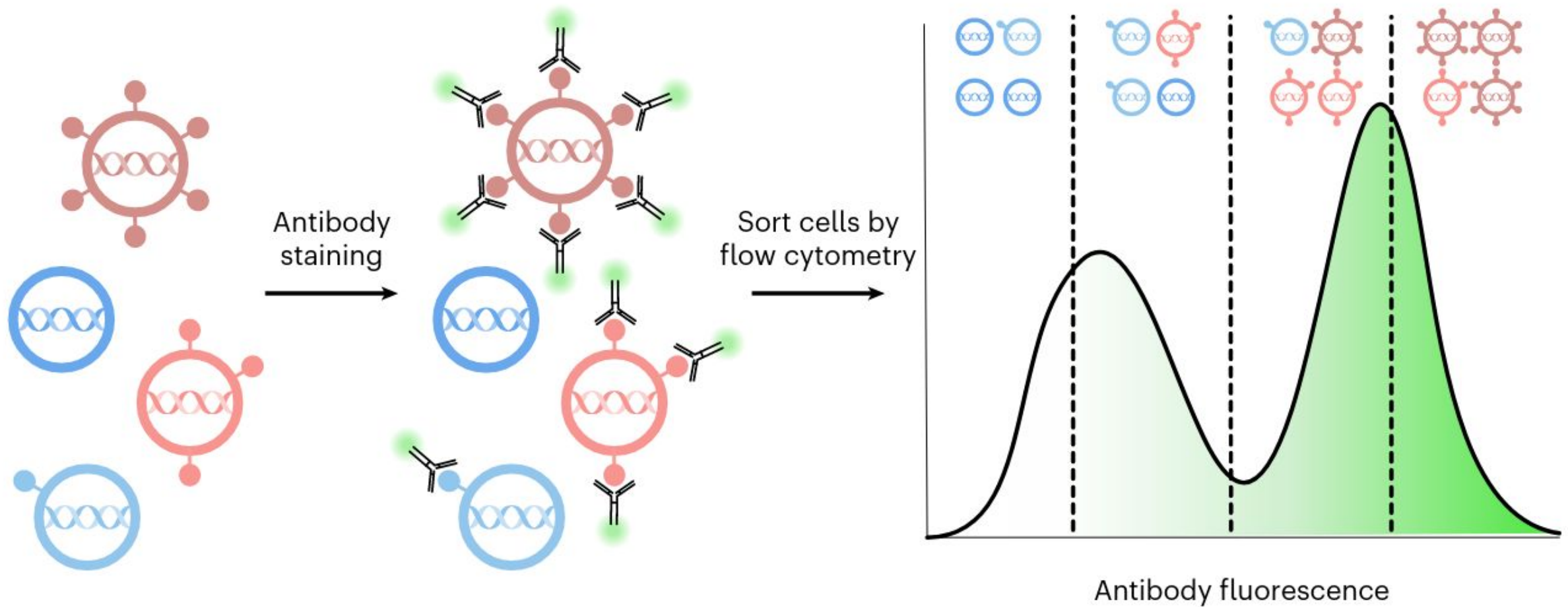
Human Domainome yeast stability experiment



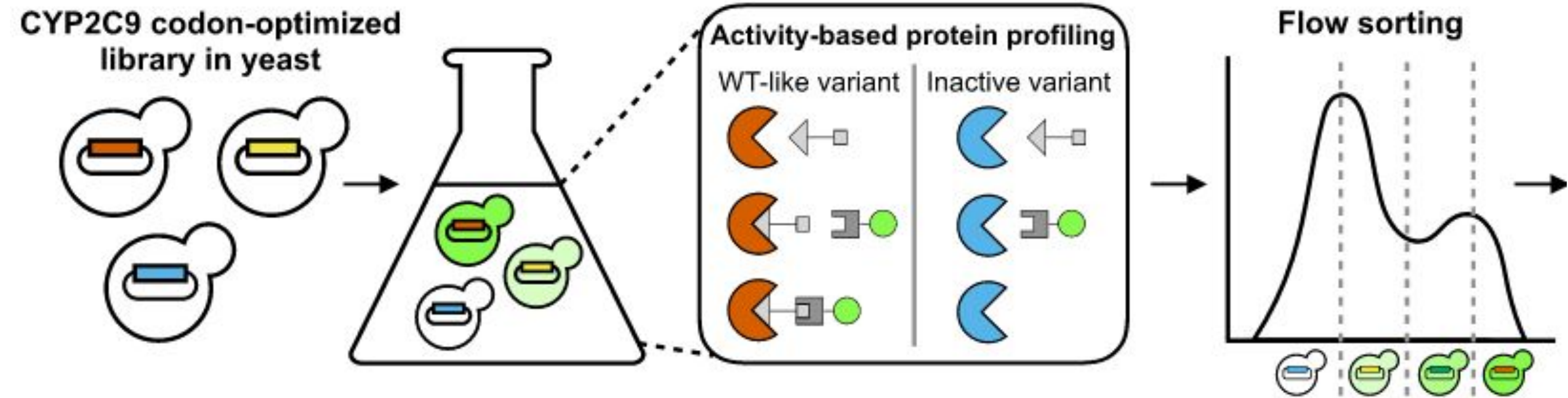
MultiSTEP: DMS for secreted proteins



MultiSTEP: DMS for secreted proteins



click-seq: profiling enzymatic activity with DMS and FACS



Experiments and data formats are heterogeneous

	A	B	C	D	E
1		sequence	name	100 nm Bcl-xL	100 nm Mcl-1
2	A1	MRPEIWIAQELRRIGDEFNAYYARRVZ	Bim	102158715.91	48726484.67
3	A2	MRPEIWIAQELRRIGDEVNAYYARRVZ	LIB2	44719093.27	101847831.9
4	A3	MRPEIWIAQELRRIGDENNAYYARRVZ	LIB3	33112634.6	109079610.26
5	A4	MRPEIWIAQELRRFGDEFNAYYARRVZ	LIB4	134182581.26	30231775.88
6	A5	MRPEIWIAQELRRFGDEVNAYYARRVZ	LIB5	36231804.96	100832032.79
7	A6	MRPEIWIAQELRRFGDENNAYYARRVZ	LIB6	34541032.83	35650206.41
8	A7	MRPEIWIAQELRRDGDEFNAYYARRVZ	LIB7	118089888.27	7233422.72
9	A8	MRPEIWIAQELRRDGDEVNAYYARRVZ	LIB8	19042519	35121970.04
10	A9	MRPEIWIAQELRRDGDENNAYYARRVZ	LIB9	13144871.33	5978852.54
11	A10	MRPEIWIAQELRRNGDEFNAYYARRVZ	LIB10	140189071.17	8712365.88
12	A11	MRPEIWIAQELRRNGDEVNAYYARRVZ	LIB11	44008865.14	58978529.28

	A	B	C	D
1	Nucleotide Position	Nucleotide Mutation	Amino Acid Substitution	RF index
2	30	A30T	Q7L	0.037281726
3	32	T32C	C8R	0.120064898
4	33	G33A	C8Y	0.869537172
5	34	C34T	C8C	0.426665334
6	35	T35C	F9L	0.11946419
7	36	T36C	F9S	0.078118763
8	36	T36A	F9Y	0.326153588
9	38	A38G	N10D	0.157902285

	A	B	C	D	E	F	G	H
1	Pos	Genotype	Env	type	DNA_1	DNA_2	virus_1	virus_2
2	1	A1C	I1L	missense	0	0	0.00033	0
3	1	A1G	I1V	missense	0.00081	0.00107	0.03002	0.02591
4	1	A1T	I1F	missense	0.00115	0.00069	0.00134	0.00109
5	2	T2A	I1N	missense	0.00037	0.00044	0.00031	0.00047
6	2	T2C	I1T	missense	0.00087	0.00075	0.00443	0.00279
7	2	T2G	I1S	missense	6.20E-05	5.01E-05	0	0
8	3	C3A	I1I	silent	0.00015	0.00011	0.00654	0.00465
9	3	C3G	I1M	missense	3.10E-05	5.01E-06	0.00074	0.00047

	A	B	C	D	E	F
1	Apparent Fitness Landscape of the MS2 Coat Protein					
2	Residue #	A	S	T	V	C
3	1	-0.18	-0.25	0.84	-0.15	Not Observed
4	2	0.44	0.67	1.14	-0.19	-4.00
5	3	-1.63	-1.18	0.83	0.49	-4.00
6	4	-1.33	-0.13	-0.90	-0.42	-0.23
7	5	0.56	0.03	-0.31	0.62	0.14
8	6	0.86	0.58	0.84	-1.39	-0.48
9	7	-1.66	-0.56	-0.82	-0.38	-0.81
10	8	0.09	-0.42	0.64	0.05	-4.00

F

A

I

R



Findable



Accessible



Interoperable



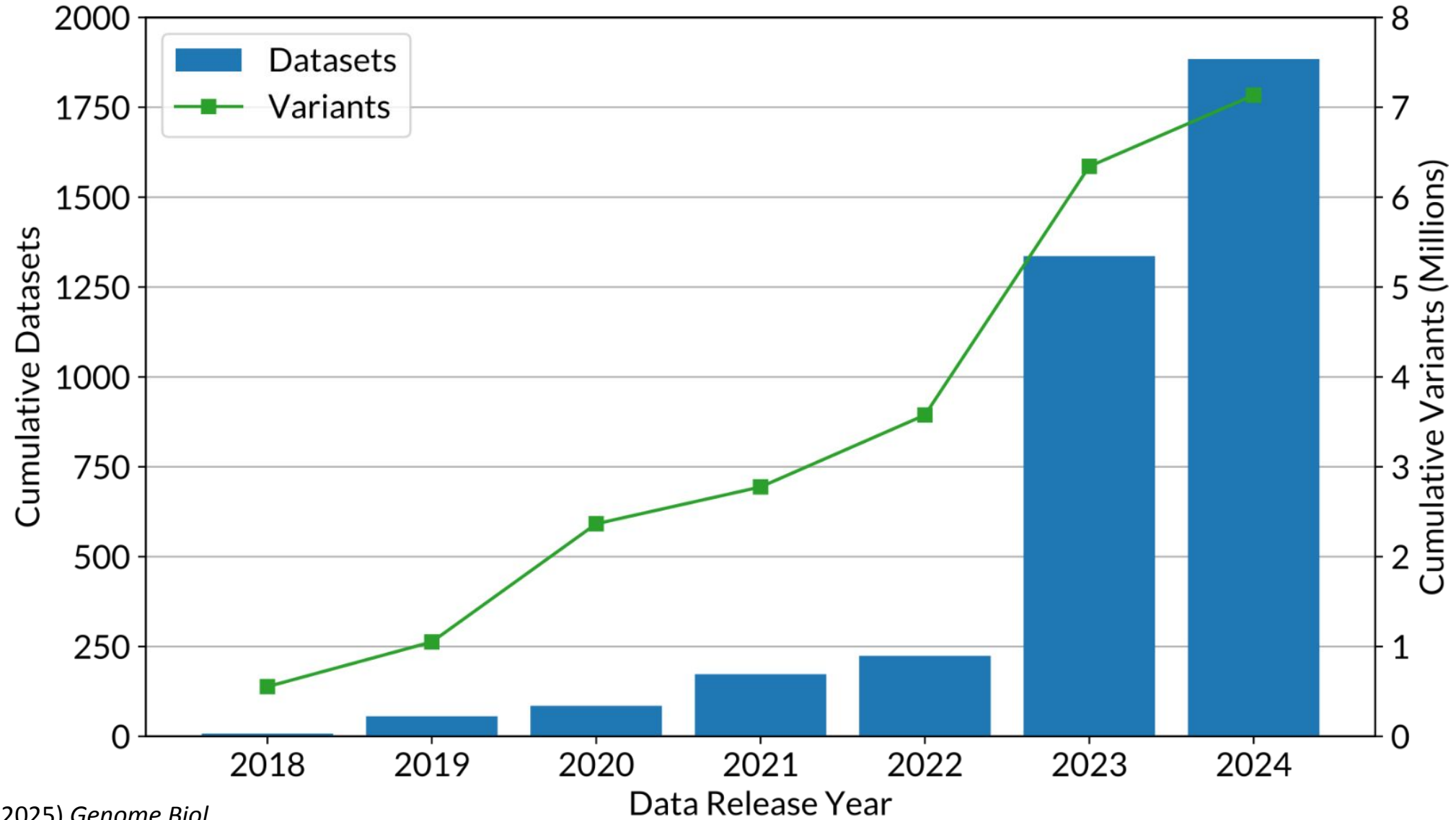
Reusable

MaveDB

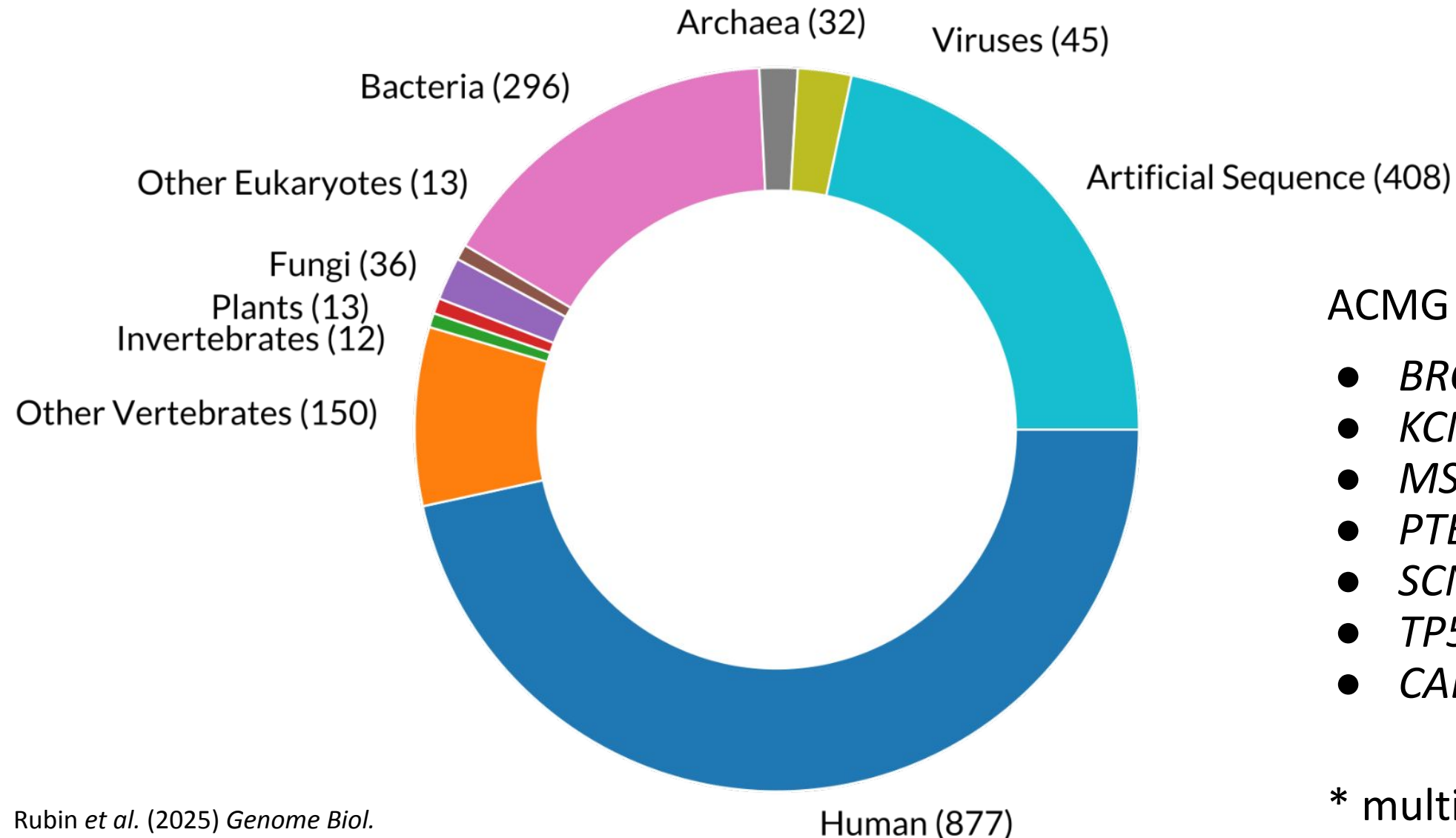
- MaveDB is the database for multiplexed functional data
- Launched in 2019
 - Esposito et al. *Genome Biol.*
- Contains ~800 datasets and 8M variant measurements
- <https://www.mavedb.org>



Millions of variants are available in MaveDB



MaveDB data are diverse but mostly human



ACMG genes in MaveDB:

- *BRCA1**
- *KCNQ1*
- *MSH2*
- *PTEN**
- *SCN5A*
- *TP53**
- *CALM1/2/3*

* multiple datasets

Hierarchical structure of MaveDB data

Experiment Set contains all assays in a study

Score Set describes the analysis and contains scores/counts

Experiment Set *urn:mavedb:00000013*

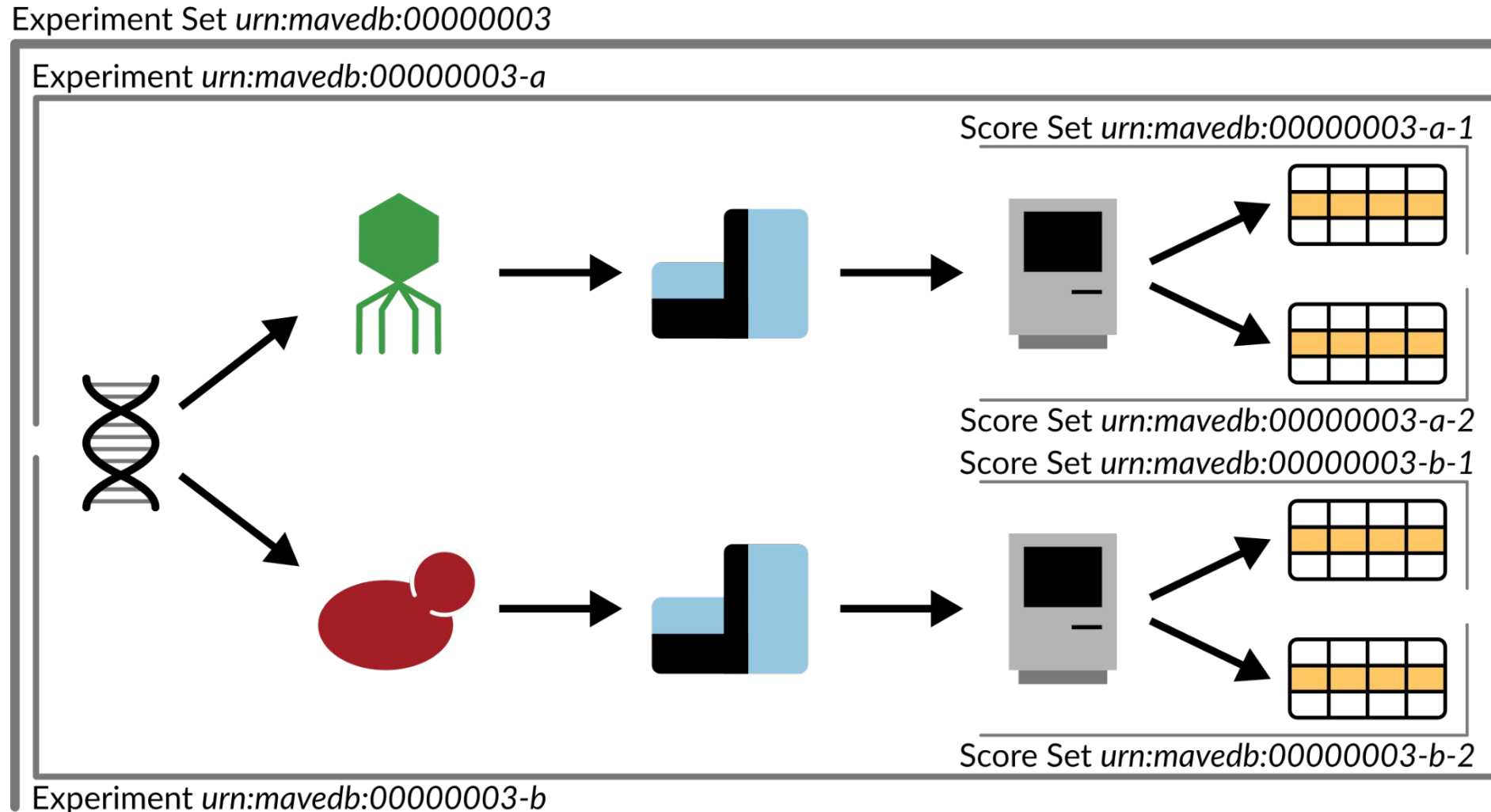
Experiment *urn:mavedb:00000013-a*

Score Set *urn:mavedb:00000013-a-1*



Experiment describes one assay and its replicates

MaveDB supports complex experimental designs



MaveDB supports reanalysis

Meta-analysis Score Set
depends on one or
more existing score sets

Experiment Set *urn:mavedb:00000013*

Experiment *urn:mavedb:00000013-a*

Score Set *urn:mavedb:00000013-a-1*



Experiment Set *urn:mavedb:00000101*

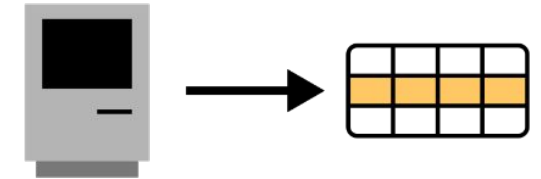
Experiment *urn:mavedb:00000101-a*

Score Set *urn:mavedb:00000101-a-1*



Experiment Set *urn:mavedb:00000102*

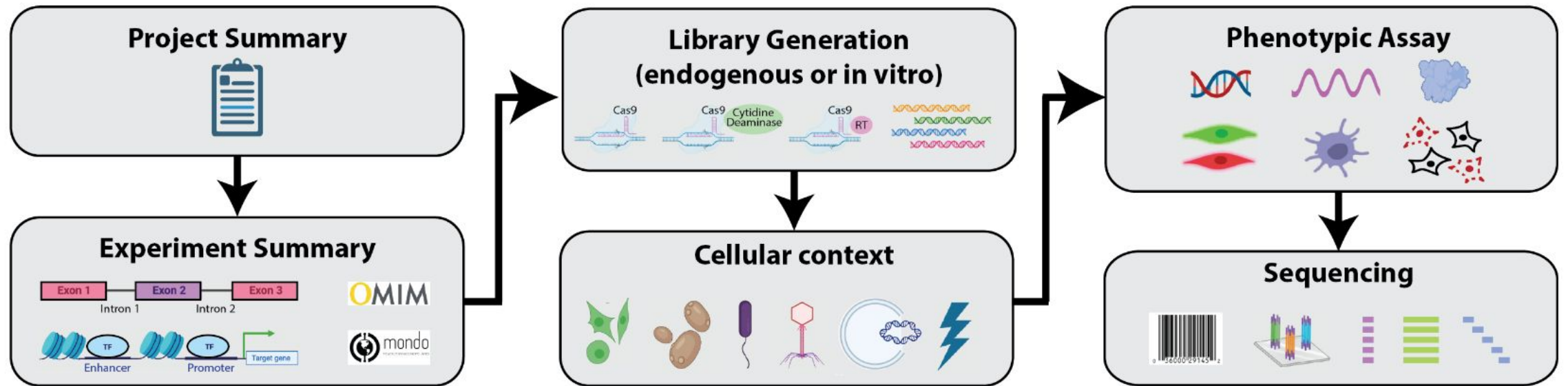
Score Set *urn:mavedb:00000102-0-1*



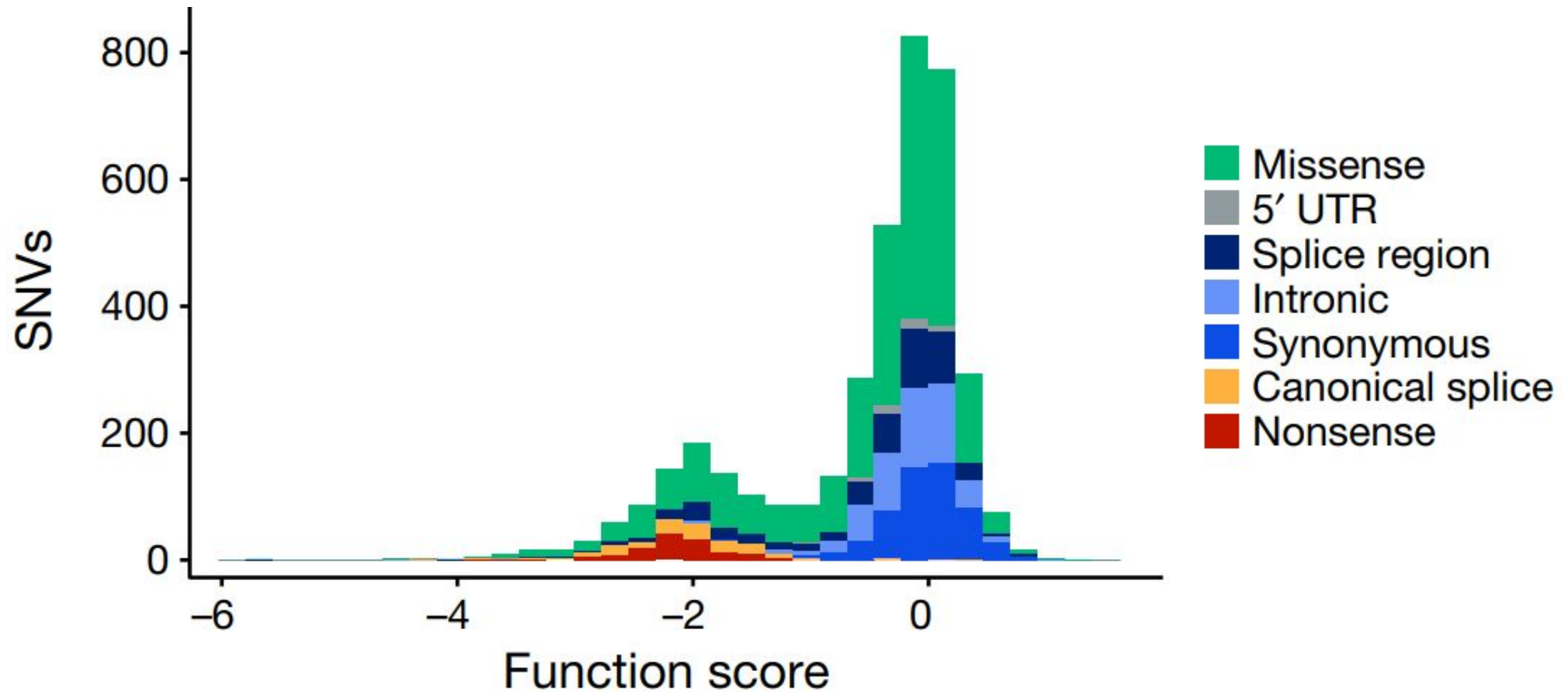
Example of what MaveDB data look like

Required	hgvs_pro	score	sd	expts	se	lower_ci	upper_ci	Optional
	p.Gly44Glu	0.849	0.040	4	0.020	0.810	0.889	
	p.His118Leu	0.382	0.272	10	0.086	0.213	0.550	
	p.His118Lys	0.313	0.195	5	0.087	0.142	0.484	
	p.Cys83Asp	1.192	0.444	2	0.314	0.578	1.807	
	p.Asp252Thr	0.047	0.103	3	0.059	-0.069	0.164	
	p.Asp52Tyr	0.901	0.118	3	0.068	0.767	1.034	

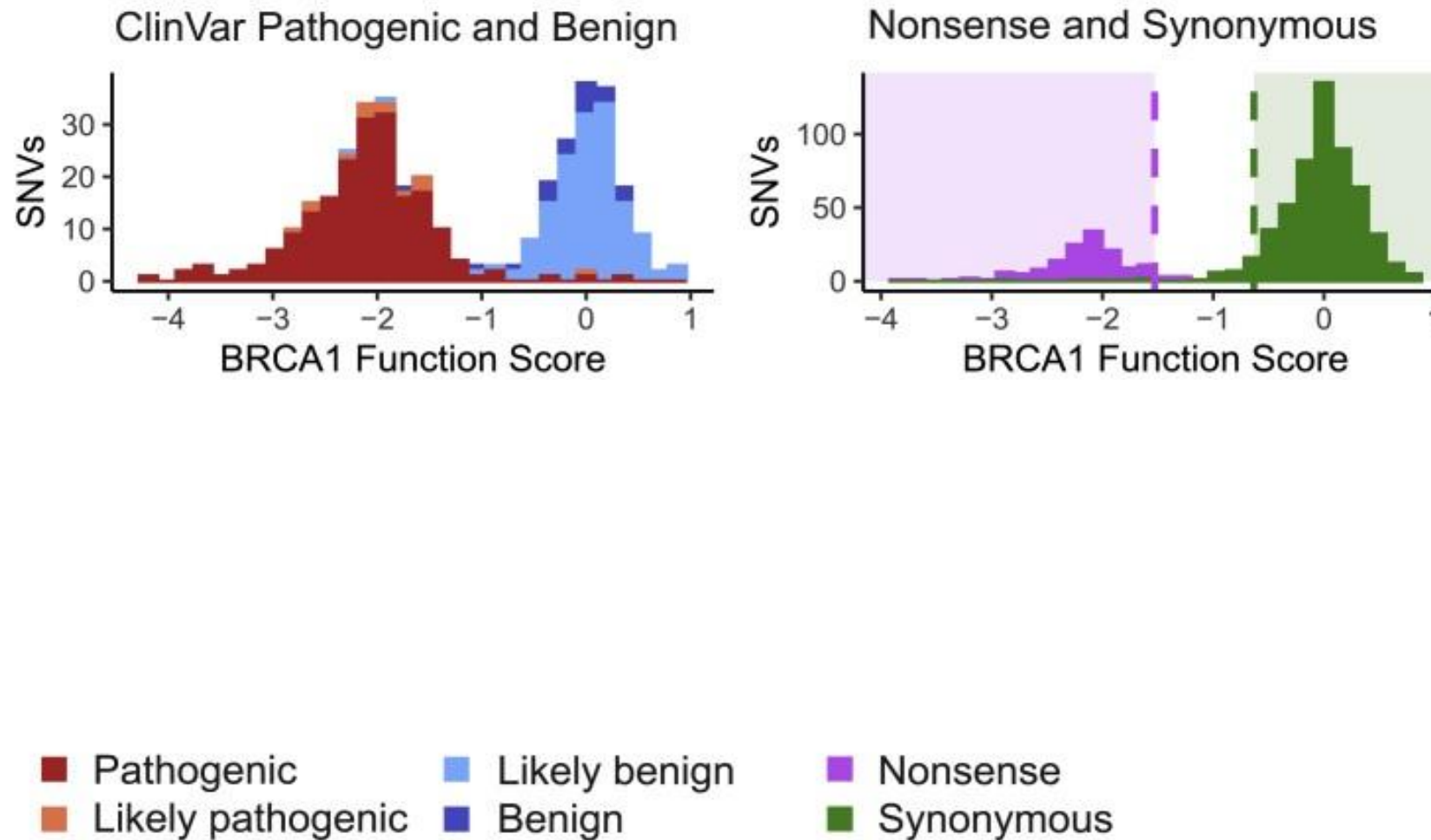
Minimum information standards and a controlled vocabulary to make sense of MAVEs



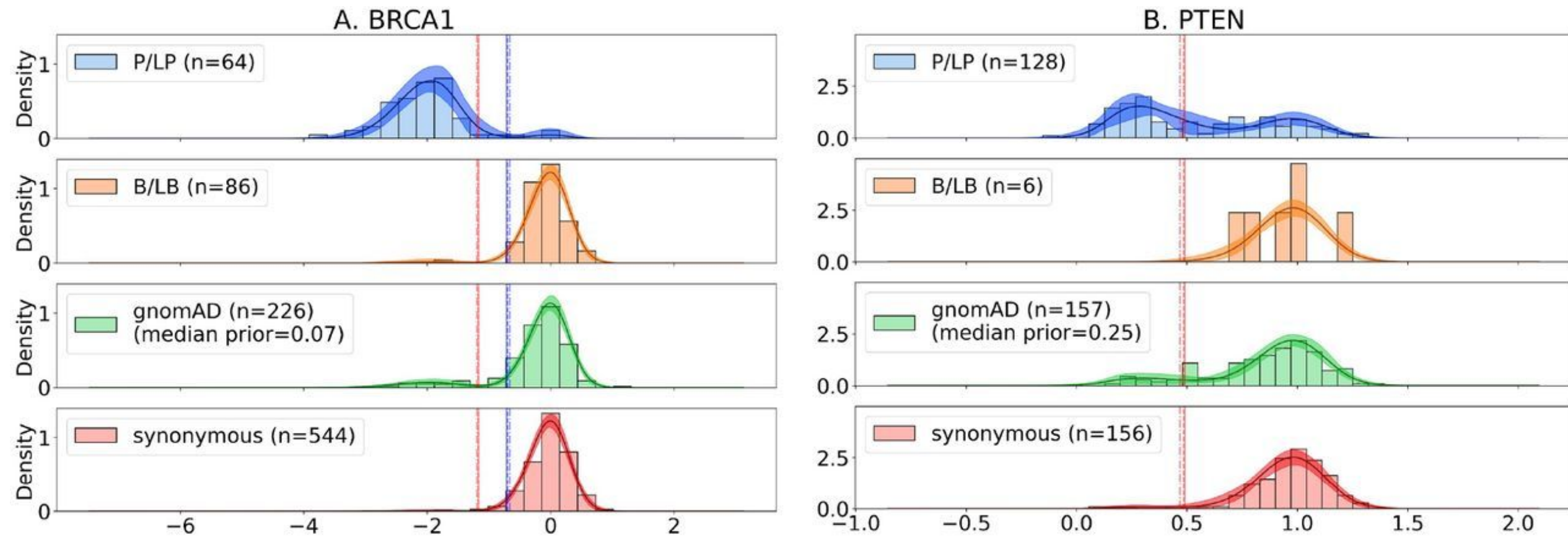
Using MAVE variants clinically



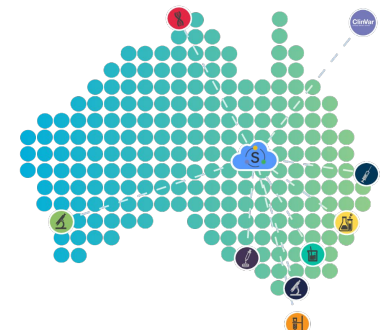
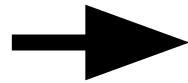
Evaluating the strength of a MAVE assay



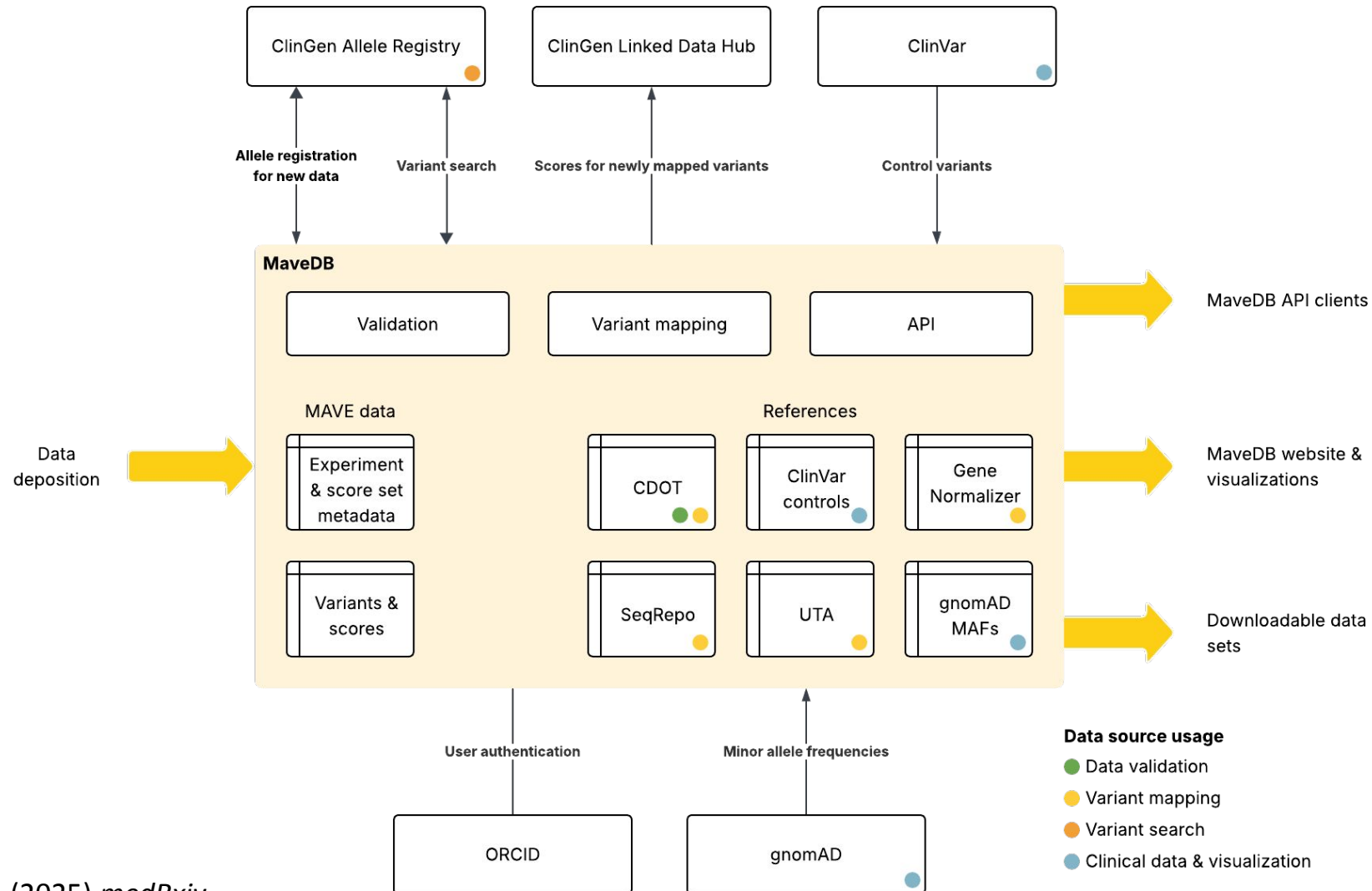
New methods for calibrating MAVEs



MaveDB is adopting and developing GA4GH standards



MaveDB data integrations



Mock-up

Prototype for functional data in ClinVar

ClinVar
Genomic variation as it relates to human health

scv001243845

Search ClinVar ?

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NM_007294.4(BRCA1):c.4945A>T (p.Arg1649Ter)

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Germline

Classification ⓘ
★ ★ ☆ ☆ (3) ⓘ

Pathogenic
criteria provided, multiple submitters, no conflicts

On this page

Classification Summary

Variant Details

Genes

Germline

Conditions

Functionally abnormal -- loss of function

assessed by cellular survival assay

this variant -1.35

non-functional

intermediate

functional

-1.328

-0.748

The saturation genome editing (SGE) assay for BRCA1
NM_007294.3:c.5520C>G, a MISSENSE variant, produced a function score of
-1.37, corresponding to a functional classification of LOSS_OF_FUNCTION.

Citation: PMID:30209399 BioRxiv

Link: MaveDB

Brotman Baty Institute University of Washington

Accession: SCV001243845.1 Submitted: Nov 12, 2018

Assay description:

saturation genome editing in haploid cells to assay the RING and BRCT domains,
which critically underlie BRCA1's role as a tumor suppressor

• Disease context: Breast-ovarian cancer, familial 1

• Transcript, Protein change: NM_007294.3:c.4945A>T, R1649

• Molecular Phenotype Measured: cellular fitness

• Cell line: human haploid cell line HAP1

• Collection Method: in vitro

• Species: human

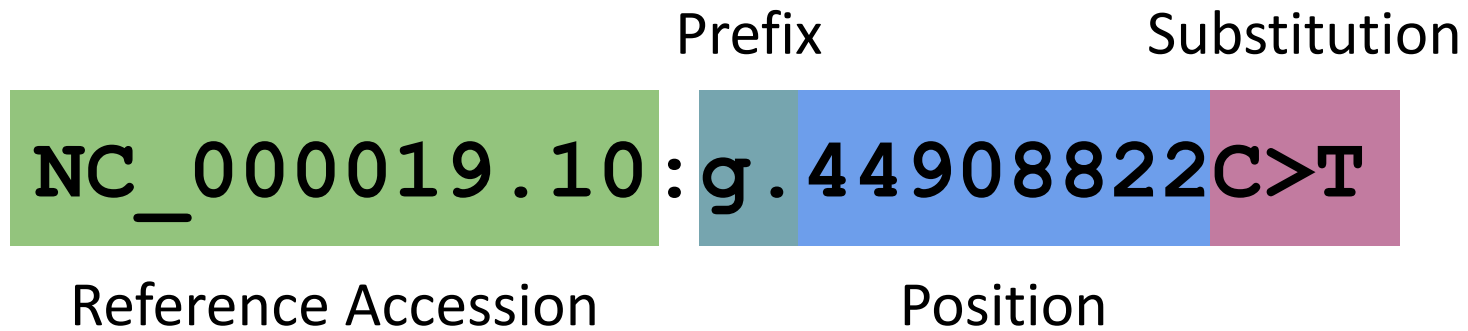
• Number of replicates: 3

• Number of controls: 3

HGVS is a common format for human variants

NC_000019.10:g.44908822C>T

Reference Accession Prefix Position Substitution



HGVS can become very complex and hard to parse

NC_000019.10:g.44908822C>T

NC_000008.11:g.(131500001_136400000)
ins[(127300001_131500000)
(131500001_136400000)inv]

GA4GH VRS is verbose but easily computable

```
{
  "id": "ga4gh:VA._YNe5V9kydfkGU0NRyCMHDSKHL4YNvc",
  "location": {
    "interval": {
      "end": {
        "type": "Number",
        "value": 44908822
      },
      "start": {
        "type": "Number",
        "value": 44908821
      },
      "type": "SequenceInterval"
    },
    "sequence_id": "refseq:NC_000019.10",
    "type": "SequenceLocation"
  },
  "state": {
    "sequence": "T",
    "type": "LiteralSequenceExpression"
  },
  "type": "Allele"
}
```

NC_000019.10:g.44908822C>T



Global Alliance
for Genomics & Health
Collaborate. Innovate. Accelerate.

Variation Representation Specification (VRS)
Wagner et al. (2021) *Cell Genom.*

Project Goal: Provide structured, machine-readable data and metadata for all functional assay results

Building GA4GH models for functional data in VA-Spec

Variant Functional Score
(Study Result)

0.26

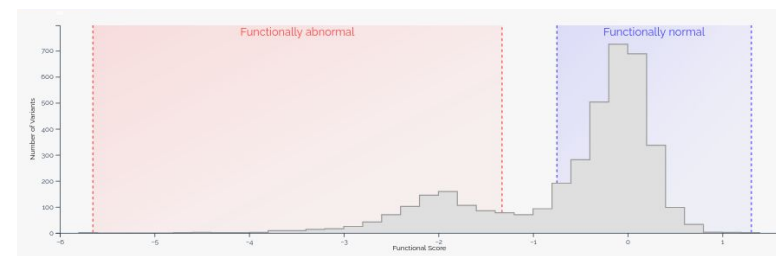
Variant Functional Class
(Statement)

normal

Variant Func. Evidence
(Evidence Line)

BS3_Strong

hgvs_pro	score	sd	expts	se	lower_ci	upper_ci
p.Gly44Glu	0.849	0.040	4	0.020	0.810	0.889
p.His118Leu	0.382	0.272	10	0.086	0.213	0.550
p.His118Lys	0.313	0.195	5	0.087	0.142	0.484



Odds Path Abnormal*	Odds Path Normal*
52.4	0.02
PS3_Strong	BS3_Strong

ExperimentalVariantFunctionalImpactStudyResult

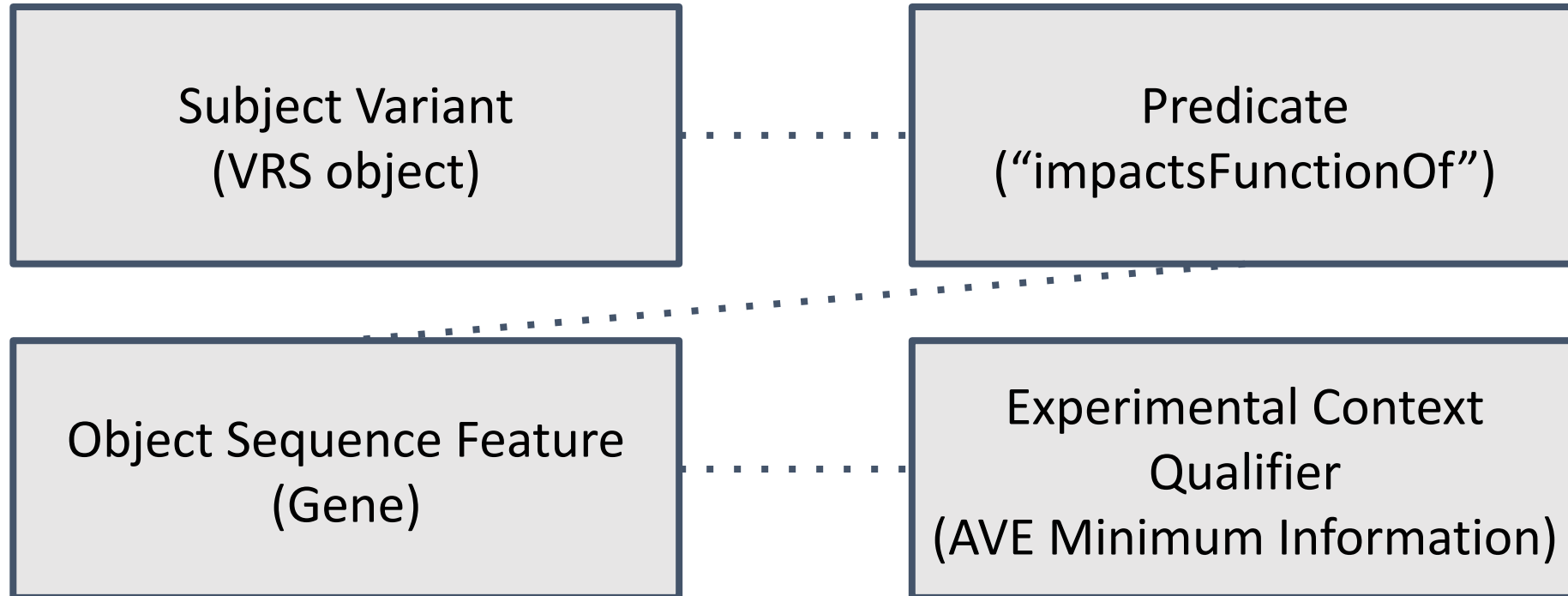
Focus Variant
(VRS object)

Functional Impact Score
(number)

Specified By
(Method - Reported In)

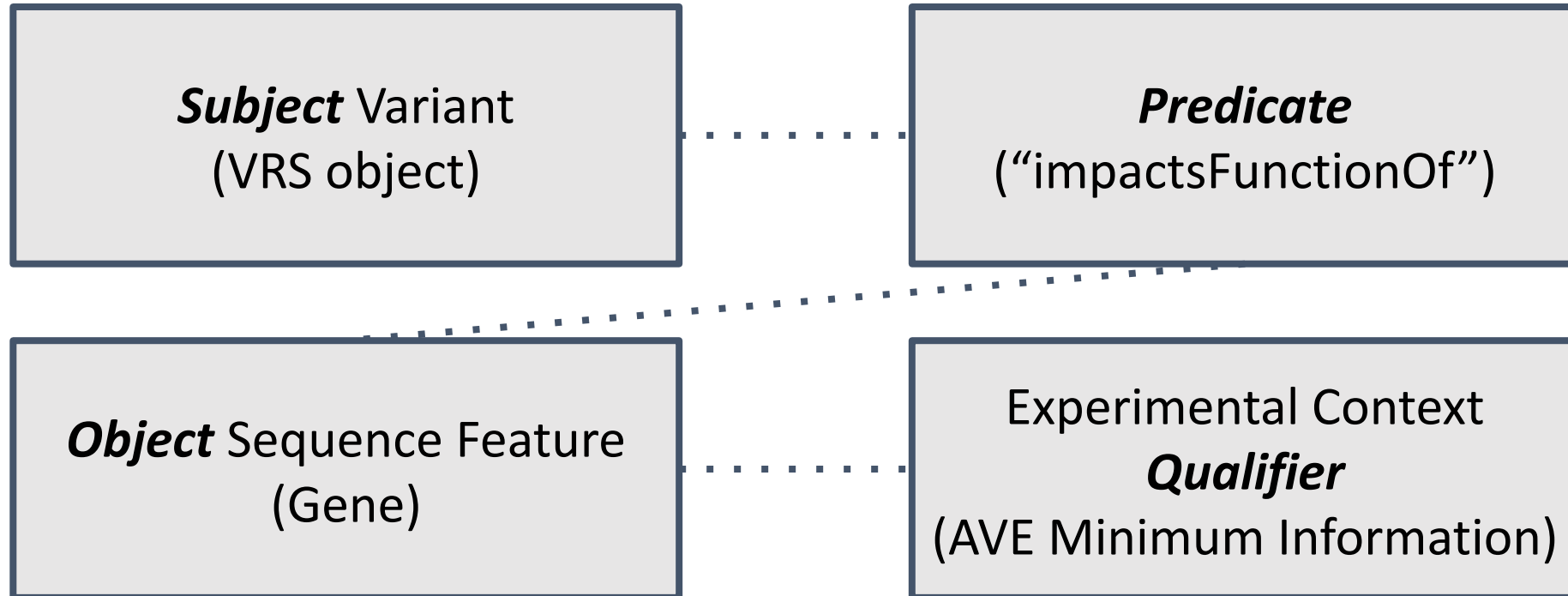
Source Data Set
(DataSet - Reported In)

ExperimentalVariantFunctionalImpactProposition



Note: The Proposition is part of a Statement, describing whether the Proposition is supported or not

ExperimentalVariantFunctionalImpactProposition



Note: The Proposition is part of a Statement, describing whether the Proposition is supported or not

Subject-Predicate-Object-Qualifier (SPOQ)

Subject Variant
(VRS object)

Predicate
("impactsFunctionOf")

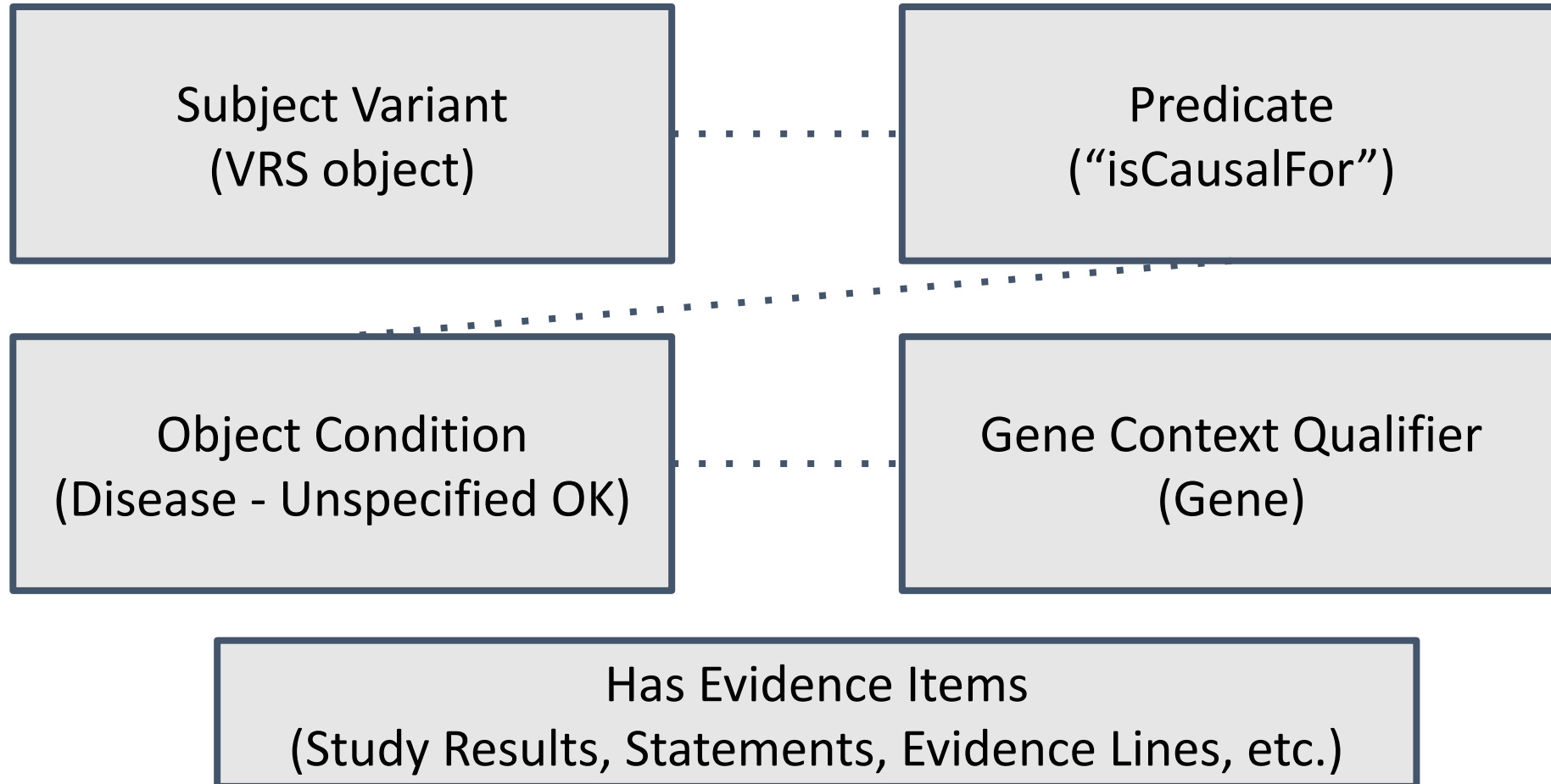
Object Sequence Feature
(Gene)

Experimental Context
Qualifier
(AVE Minimum Information)

“Variant X impacts function of Gene Y according
to Experimental Assay Z”

Note: The Proposition is part of a Statement, describing
whether the Proposition is supported or not

VariantPathogenicityProposition



Note: The Proposition is part of a Statement, describing whether the Proposition is supported or not and providing strength (including ACMG code)

Subject-Predicate-Object-Qualifier (SPOQ)

Subject Variant
(VRS object)

Predicate
("isCausalFor")

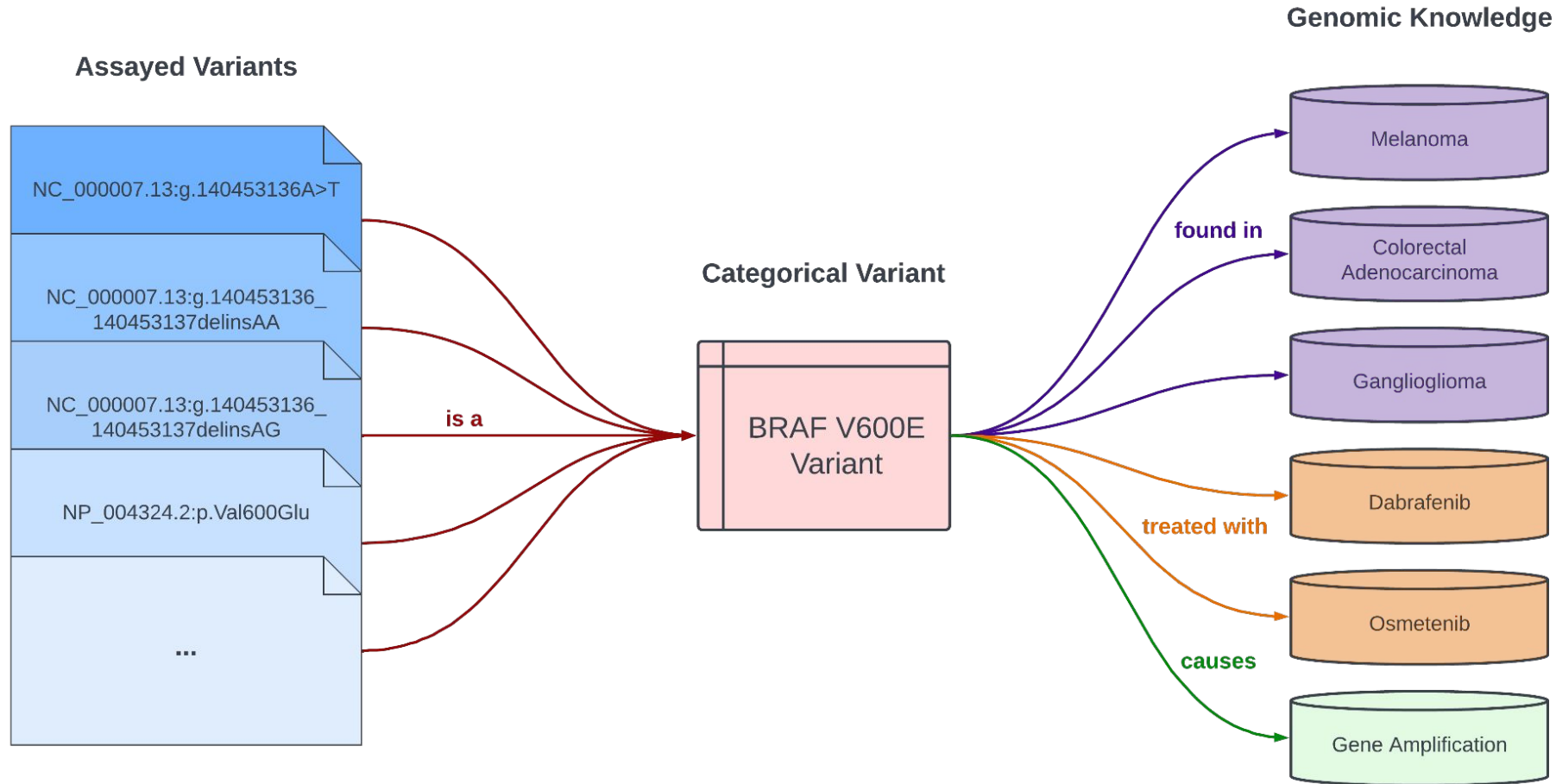
Object Condition
(Disease - Unspecified)

Gene Context **Qualifier**
(Gene)

“Variant X is causal for Unspecified Disease by impacting Gene Y”

Note: The Proposition is part of a Statement, describing whether the Proposition is supported or not

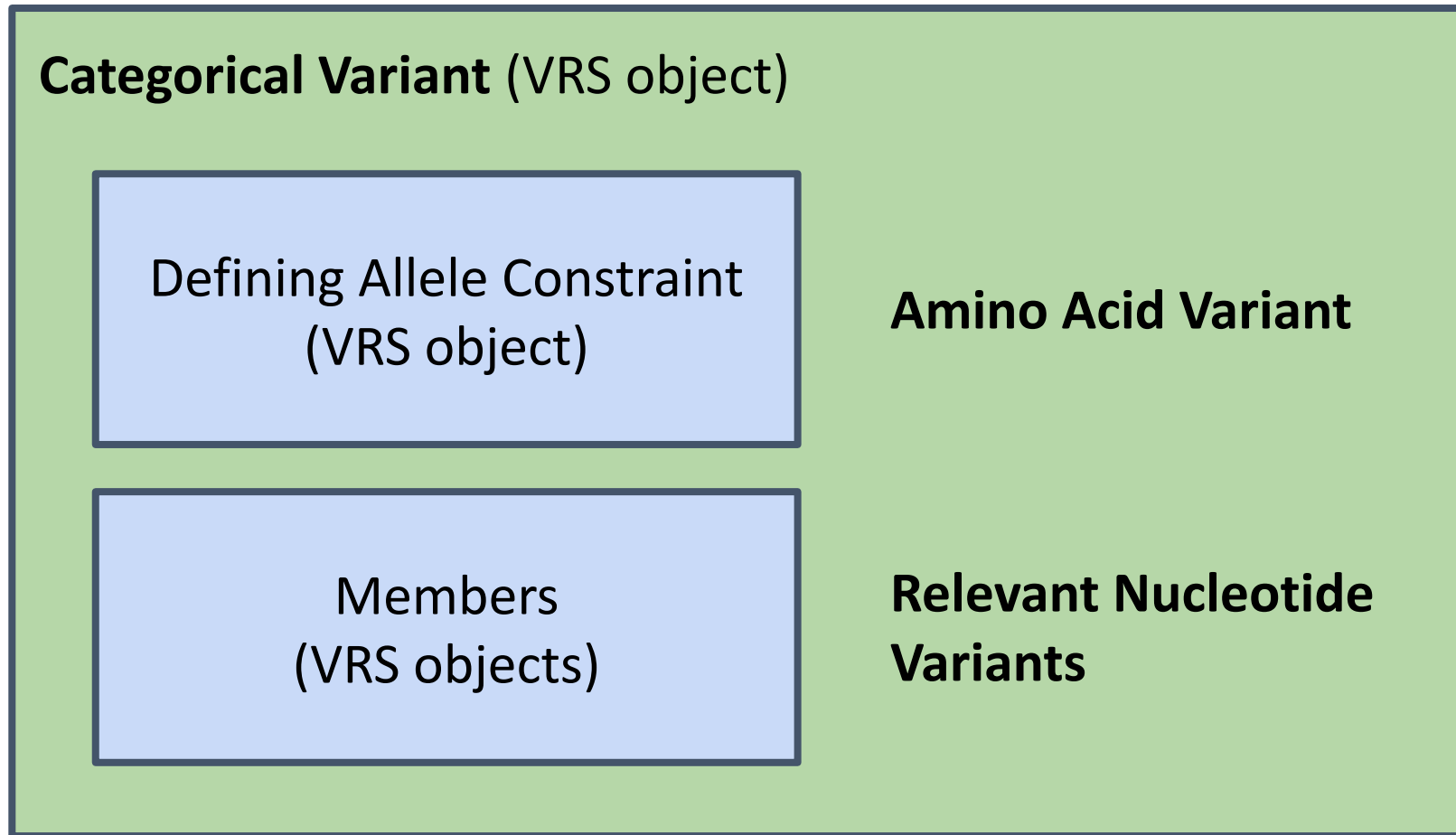
What is categorical variation?



Cat-VRS is a GA4GH standard for categorical variation



Using Cat-VRS to represent amino acid variants



Uploading data to MaveDB – <https://staging.mavedb.org>

MaveDB (Staging) x +

https://staging.mavedb.org

MAVEDB Search

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MAVEMD Looking for the new MaveMD clinical features? [Click here.](#)

About

MaveDB is a public repository for datasets from Multiplexed Assays of Variant Effect (MAVEs), such as those generated by deep mutational scanning (DMS) or massively parallel reporter assay (MPRA) experiments.

MaveDB is open-source, released under the [AGPLv3](#) license.

MaveDB is hosted by the [Fowler Lab](#) in the [Department of Genome Sciences](#) at the University of Washington. It is supported and developed by the [University of Washington](#), the [Walter and Eliza Hall Institute of Medical Research](#), and the [Brotman Baty Institute](#).

If you have questions, comments, or suggestions, check out the [Help & Support](#) page for information about how to get in touch.

Featured Searches

Organisms	Targets
Homo sapiens Mus musculus Saccharomyces cerevisiae S288C	Homo sapiens KCNQ4 TEM-1 β-lactamase

Signing in to MaveDB

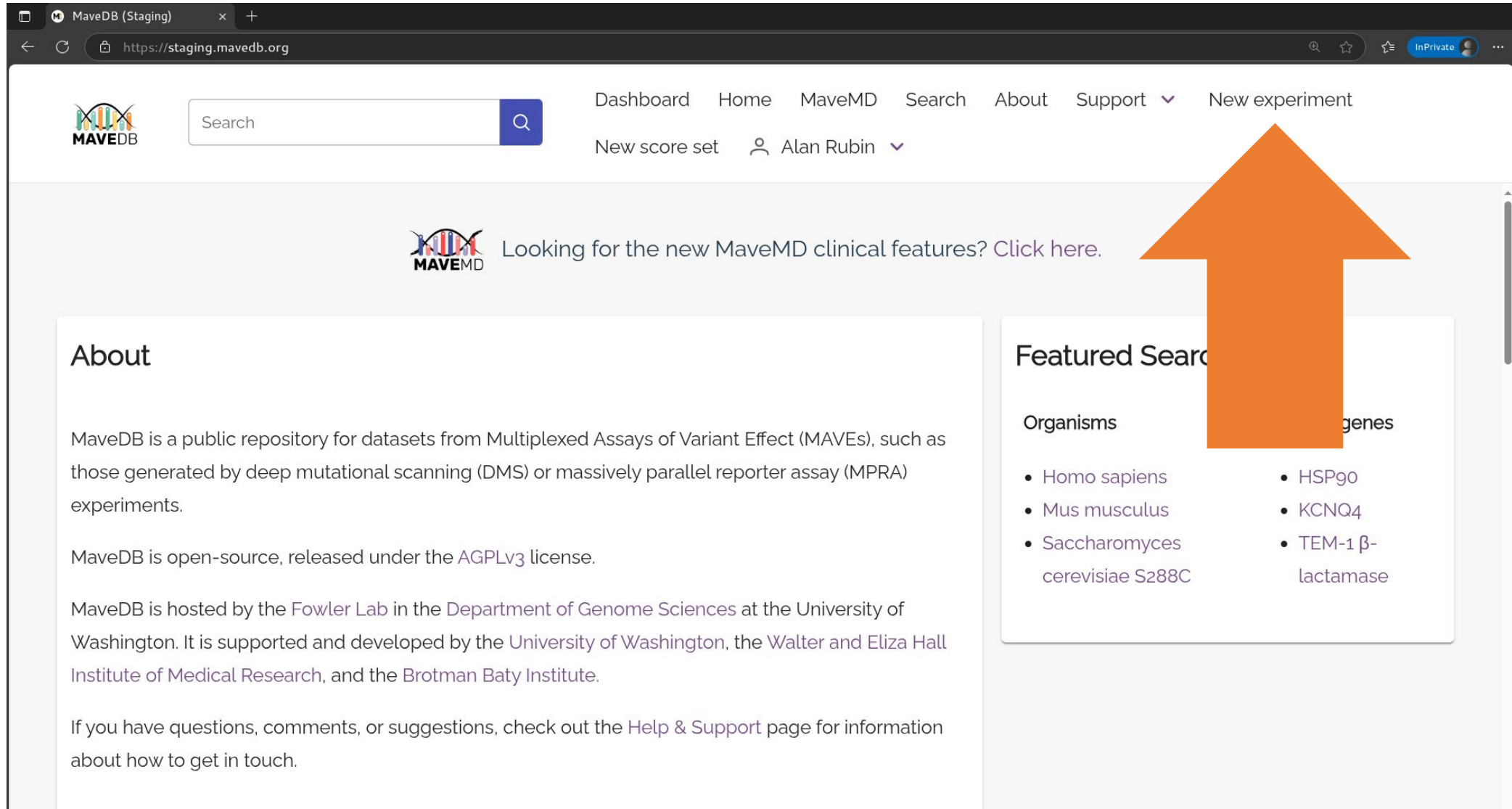


- MaveDB authentication is performed through ORCID
- MaveDB uses this information to display your name, etc.

If you do not have an ORCID ID please create one!

<https://orcid.org>

Creating a new Experiment



The screenshot shows the MaveDB (Staging) website interface. The top navigation bar includes links for Dashboard, Home, MaveMD, Search, About, Support, and New experiment. A large orange arrow points to the 'New experiment' link. Below the navigation bar, there is a search bar and a user profile section for Alan Rubin. The main content area features an 'About' section on the left and a 'Featured Search' section on the right. The 'About' section describes MaveDB as a public repository for datasets from Multiplexed Assays of Variant Effect (MAVEs), such as those generated by deep mutational scanning (DMS) or massively parallel reporter assay (MPRA) experiments. It also mentions that MaveDB is open-source, released under the AGPLv3 license, and is hosted by the Fowler Lab in the Department of Genome Sciences at the University of Washington. The 'Featured Search' section lists organisms and genes for search.

MaveDB (Staging)
https://staging.mavedb.org

Dashboard Home MaveMD Search About Support **New experiment**

New score set Alan Rubin

MAVEDB Looking for the new MaveMD clinical features? [Click here.](#)

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Featured Search

Organisms	Genes
Homo sapiens Mus musculus Saccharomyces cerevisiae S288C	HSP90 KCNQ4 TEM-1 β-lactamase

MaveDB example dataset

- Dataset is from G6PD VAMP-seq
- Do your best to fill out the metadata based on what you have learned so far in the course!
- After creating the experiment, go ahead and add a new score set
 - The score file and target sequence are available in the GitHub data folder
 - Try creating thresholds after the variants populate

Exploring MaveMD data

MaveDB | MaveMD varia x +

← ↻ 🔒 https://www.mavedb.org/mavemd?searchType=hgvs

☆ ⚙ InPrivate



[Home](#) [MaveMD](#) [Search](#) [About](#) [Support](#) [Sign in](#)



MaveMD (MAVEs for MeDicine) is an interface that integrates ClinVar and the ClinGen Allele Registry, displays clinical evidence calibrations, provides intuitive visualizations, and exports structured evidence compatible with ACMG/AMP variant classification guidelines. MaveMD currently contains 438,318 variant effect measurements mapped to the human genome from 74 MAVE datasets spanning 32 disease-associated genes.



Search MaveDB for human gene variants

HGVS ▾

Search

×

Don't have a versioned reference sequence identifier? Click here to perform a fuzzy search instead: [Fuzzy Search](#)

[Show search examples](#)

Gene	Score set	Publication	Calibrations w. evidence / Total
ASPA	Aspartoacylase (ASPA) cellular abundance	Grønbæk-Thygesen M et al. ASPA 2024	2 / 4
	Aspartoacylase (ASPA) cellular toxicity	Grønbæk-Thygesen M et al. ASPA 2024	3 / 4
BRCA1	BRCA1 SGE Normalized Scores	Findlay et al. BRCA1 2018	5 / 6
	Scores from growth assay of BRCA1 variants	Adamovich AI et al. BRCA1 2022	5 / 5
	Scores from multiplexed functional assay of BRCA1 variants	Adamovich AI et al. BRCA1 2022	5 / 5
BRCA2	Scores from arrayed screen of BRCA2 homology directed repair function in VC-8 cells	Hu C et al. BRCA2 2024	4 / 5
CALM1, CALM2, and CALM3	Human Calmodulin DMS-TileSeq	Weile et al. CALM1 2017	1 / 2

Looking up a variant – ClinVar

<https://www.ncbi.nlm.nih.gov/clinvar/variation/182595/>

ClinVar Genomic variation as it relates to human health

Search by gene symbols, location, HGVS expressions, c-dot, p-dot, conditions, i

Search ClinVar ?

Advanced search

About Access Submit Stats FTP Help



NM_000251.3(MSH2):c.55T>C p.Phe19Leu Cite Follow Print Download

 We've updated the full submission template; previous versions are no longer supported. Download the latest version from [the ClinVar FTP site](#) and [read more about the available templates.](#)

Germline

Classification
★☆☆☆ ?
Conflicting classifications of pathogenicity
Uncertain significance(2); Benign(4); Likely benign(9)
15 out of 17 submissions contributed to this classification ?



Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity



Looking up a variant – ClinGen Allele Registry



ClinGen Allele Registry

[Allele Registry](#) | [Register](#) | [Login](#) | [Forgot Login?](#)

Canonical Allele Identifier: PA299391

Gene: MSH2 [HGNC](#) [NCBI](#)

Linked Data - NCBI & NCI

ClinVar RCV:

[RCV000160635](#) [RCV000409531](#)
[RCV000588459](#) [RCV001084278](#)
[RCV001255213](#) [RCV003323295](#)
[RCV003492654](#) [RCV004528902](#)

ClinVar Variation:

[182595](#)

Linked Data - Variant Effect Evidence

MaveDB:

Score Set: [urn:mavedb:00000050-a-1#353](#)

Score: -3.7623108789

Score-Set Keywords (AI) ⓘ:

human HAP1 cells

LOF

MSH2 loss-of-function

[JSON-LD](#) ⓘ

Looking up a variant – exploring available datasets



The screenshot shows the MaveMD website interface. The browser address bar displays `https://www.mavedb.org/mavemd?searchType=hgvs`. The website header includes a search bar, navigation links (Home, MaveMD, Search, About, Support, Sign in), and a user profile icon labeled "InPrivate".

The main content area features the MaveMD logo and a description: "MaveMD (MAVEs for MeDicine) is an interface that integrates ClinVar and the ClinGen Allele Registry, displays clinical evidence calibrations, provides intuitive visualizations, and exports structured evidence compatible with ACMG/AMP variant classification guidelines. MaveMD currently contains 438,318 variant effect measurements mapped to the human genome from 74 MAVE datasets spanning 32 disease-associated genes."

Below the description is a search section titled "Search MaveDB for human gene variants". It includes a dropdown menu set to "HGVS", a search input field with the placeholder "Enter a value", a "Search" button, and a "Fuzzy Search" button. A link "Don't have a versioned reference sequence identifier? Click here to perform a fuzzy search instead:" is also present.

A link "Show search examples" is located below the search section. Below this link is a table of search results:

Gene	Score set	Publication	Calibrations w. evidence / Total
ASPA	Aspartoacylase (ASPA) cellular abundance	Grønbæk-Thygesen M et al. ASPA 2024	2 / 4
	Aspartoacylase (ASPA) cellular toxicity	Grønbæk-Thygesen M et al. ASPA 2024	3 / 4
BRCA1	BRCA1 SGE Normalized Scores	Findlay et al. BRCA1 2018	5 / 6
	Scores from growth assay of BRCA1 variants	Adamovich AI et al. BRCA1 2022	5 / 5
	Scores from multiplexed functional assay of BRCA1 variants	Adamovich AI et al. BRCA1 2022	5 / 5
BRCA2	Scores from arrayed screen of BRCA2 homology directed repair function in VC-8 cells	Hu C et al. BRCA2 2024	4 / 5
CALM1, CALM2, and CALM3	Human Calmodulin DMS-TileSeq	Weile et al. CALM1 2017	1 / 2

